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A Three-Pillar Framework Unifying Structural, Adaptive, and Semantic Reasoning for Graph Fraud Detection

Fady Tarek[1], Prof Dr Ayman ElBadawy[1]

[1]Department of Mechatronics Engineering, German University in Cairo, Cairo, Egypt

Abstract

Graph neural networks are a standard substrate for fraud detection on multi-relation review graphs, yet sophisticated fraudsters evade neighbourhood aggregation by camouflaging both node attributes and connections. Existing defences are partial: reinforcement-learning approaches such as CARE-GNN rely on heuristic bandits with limited per-node adaptability and no external signal, while recent GNN–LLM integrations enrich node features but lack a learned policy for dynamic relation weighting. We propose a three-pillar framework unifying structural, adaptive, and semantic reasoning under a single joint objective. The structural pillar is a multi-relation GNN backbone—instantiated with CARE-GNN, though not limited to it; the adaptive pillar, the Camouflage-Aware Policy Network, replaces the heuristic bandit with an actor–critic policy that learns per-node, per-relation filtering thresholds; the semantic pillar injects LLM-derived reasoning scores into the policy state and fraud-specific textual risk scores into node features via a dimensionality-preserving gated residual. On YelpChi the full framework improves on its structural base by +2.70 pp AUC across twenty held-out seeds, and by +1.80 pp across ten independent data splits (10/10 positive, $p = 3 \times 10^{-6}$). We report both, because repeated seeds on a fixed split measure only initialisation variance—which we find is 2.4–2.8× smaller than the split-to-split variance that a generalisation claim actually rests on. Both learned pillars contribute significantly and compose, and the adaptive pillar replicates on a second dataset. The semantic signal generalises well beyond the framework: concatenated onto eight backbones spanning linear models, tree ensembles, and relational and spectral GNNs, it significantly improves seven, including the tree-plus-aggregation method that currently leads this benchmark, and a second selection-based GNN by +6.58 pp. We further show that the pillar's value lies in *which* six quantities are scored rather than in who scores them—a deterministic, zero-cost scorer matches the language model on every graph backbone tested—so the signal is reproducible with no API dependence.

Keywords: Graph neural networks; Fraud detection; Reinforcement learning; Large language models; Multi-relation graphs

1 Introduction

Modern digital systems generate fraud at scale: coordinated groups of malicious actors manipulate financial transactions, exploit platform incentives, and fabricate social signals, distorting trust and inflicting economic harm across domains [1]. Fraud is fundamentally relational—perpetrators act in groups, share targets, and reuse behavioural patterns—so graph neural networks (GNNs) are a natural substrate for detection. By modelling entities and their interactions as multi-relation graphs, GNNs can learn discriminative representations from both attributes and connectivity [2].

In response, fraudsters deliberately evade neighbourhood aggregation. They alter their attributes to resemble legitimate users and connect to benign entities to dilute the suspiciousness of their neighbourhoods—*relation camouflage* [3]. Defending against camouflage requires three complementary capabilities: structural reasoning to expose relational signatures that camouflage tries to hide; adaptive reasoning to decide, per node and per relation, which neighbours to trust; and semantic reasoning to incorporate fraud signals beyond the behavioural feature space that graph structure alone can capture.

Prior work typically addresses these capabilities in isolation. CARE-GNN [3] introduced reinforcement-learning-guided neighbour selection, but uses a heuristic Bernoulli bandit with a single global thresh-

old per relation and limited ability to adapt per node or incorporate external signals. RioGNN [4] refined this idea with recursive multi-relation policies, but still relies on a largely structural state representation. Recent GNN–LLM integrations for fraud detection—notably FLAG [5]—inject LLM-derived signals into node features and sampling decisions, but do not learn a policy that dynamically weights relations at the node level. To the best of our knowledge, no prior work unifies structural, adaptive, and semantic reasoning within a single end-to-end framework.

We argue that such unification is necessary under camouflage: structural reasoning alone can be misled by adversarial connectivity, adaptive reasoning without external signals can only refine what the structural layer already observes, and semantic reasoning without relational grounding can miss the coordinated structure that defines fraud rings. We instantiate this unified design and, through a pillar-wise ablation, quantify how much each mode of reasoning contributes—finding every pillar individually significant under held-out evaluation, the semantic pillar largest, and the two learned pillars composable once a suppressive coupling between them is removed (Section 4).

We propose a three-pillar framework to close this gap. The structural pillar uses a multi-relation GNN backbone (we instantiate it with CARE-GNN [3]). The adaptive pillar replaces CARE-GNN's heuristic bandit with the Camouflage-Aware Policy Network (CAPN), an actor–critic policy that outputs per-node, per-relation filtering thresholds conditioned on a rich state vector. The semantic pillar leverages a large language model (LLM) to compute (i) structural-reasoning scores that enrich the policy state and (ii) fraud-specific textual risk scores that are injected into node features via a gated residual module that preserves feature dimensionality. All LLM outputs are precomputed and cached, making training cost independent of the external model. The three pillars are trained end-to-end under a single joint objective.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 details the three-pillar framework. Section 4 reports experiments and ablations. Section 5 concludes.

2 Related Work

We review three lines of work that directly motivate our contribution: GNN-based fraud detection, reinforcement learning for graph-based filtering, and LLM integration with graph learning. We focus on the specific limitation each line leaves unaddressed.

2.1 GNN-Based Fraud Detection

Graph neural networks extended representation learning to relational data by iteratively aggregating neighbourhood information, with GCN [6], GraphSAGE [7], and GAT [8] establishing the foundational aggregation mechanisms. Their direct application to fraud detection, however, is undermined by camouflage: fraudsters strategically connect to benign nodes so that standard message passing aggregates predominantly legitimate neighbour signals, suppressing the fraud signal of the target node [9].

Purpose-built architectures address this. GraphConsis [9] filters neighbours whose features are inconsistent with the centre node's predicted class. CARE-GNN [3] formalised two camouflage types—feature camouflage and relation camouflage—and introduced label-aware similarity scoring with a heuristic bandit for adaptive neighbour selection across multiple relations. PC-GNN [10] improved minority-class recall through label-balanced neighbourhood sampling. Production deployments such as xFraud [11] have demonstrated that heterogeneous GNNs scale to graphs with billions of edges through mini-batch sampling and embedding caching. A parallel spectral line—BWGNN [12], GHRN [13], and GAGA [14]—leads the GADBench benchmark [15] on YelpChi and Amazon, and offers a genuinely different account of why these graphs are hard: the difficulty is the spectral energy distribution under heterophily, not an arms race in neighbour selection. That account is well supported, and our own reimplementation confirms its strength at our split (Section 4.3). The two accounts are not exclusive. The spectral line reshapes *how* a fixed neighbourhood is filtered in the frequency domain; the selection line changes *which* neighbours enter it at all. We work in the second because it is the one that admits a per-node learned policy—and we show in Section 4.3 that our semantic signal improves the spectral models too.

These methods advance structural reasoning over relational fraud graphs, but their adaptation mechanisms remain coarse: filtering thresholds are global

rather than per-node, and no method incorporates external semantic signal into its filtering decisions.

2.2 Reinforcement Learning for Graph-Based Filtering

The neighbour filtering decision is naturally framed as a policy problem, motivating the use of reinforcement learning. CARE-GNN [3] formulated it as a multi-armed bandit: a single threshold per relation updated each epoch by a sign-based reward derived from inter-epoch changes in average label-aware distance. This produces a non-differentiable, stateless policy with three key limitations: thresholds are global across all nodes within a relation; rewards are binary, discarding magnitude information; and the policy has no interface for incorporating signals beyond its own internal distance metric.

RioGNN [4] generalised this to a recursive RL framework operating across multiple relation types simultaneously, improving both granularity and stability of the learned selection policy. However, the policy state remains structural — derived from graph statistics and label distances only — and provides no mechanism for integrating semantic signals into filtering decisions. More recent RL-based detectors continue this structural-only trend: FraudGNN-RL [16] uses a deep Q-network to adapt detection thresholds on dynamic transaction graphs, and a reinforced neighbour selector [17] learns which neighbours to trust for unsupervised graph anomaly detection—both conditioning the policy on graph statistics alone, with no semantic or language signal in the state.

Across this line the quality of the policy state is what bounds the policy: RioGNN [4] attributes much of its gain over CARE-GNN's bandit to a richer state, and the agent can only act on what that state encodes. To our knowledge, no prior work in graph fraud detection uses an actor-critic policy with a state vector designed to admit external semantic signals.

2.3 Large Language Models for Fraud Detection

LLMs have emerged as a source of semantic signal complementary to graph topology in settings where entity descriptions carry anomaly information invisible to structural analysis alone [5]. Fraudsters who successfully mimic legitimate graph structure may still exhibit detectable linguistic patterns; conversely, text-only approaches miss the relational structure that defines coordinated fraud rings.

FLAG [5] is the closest prior work. It uses an LLM to extract discriminative features from entity descriptions, applying them both as supplementary node features for the GNN and to guide neighbour sampling via LLM-based semantic similarity. Deployed on Alipay's credit risk system, FLAG demonstrated average improvements of approximately 3% in F_1-macro and 7% in AUC over competitive GNN baselines, confirming that LLM-derived semantic features provide complementary signal beyond structural aggregation.

FLAG and related integrations treat the LLM as a feature encoder: it produces representations that augment the input space or guide retrieval, but does not participate in adaptive filtering decisions during training. No prior work routes LLM-derived signals through both feature space and policy state, or conditions a learned filtering policy on semantic reasoning scores. A wave of 2025 GNN–LLM detectors follows the same encoder paradigm—DGP [18] prompts graph-enhanced LLMs at dual granularity, and multi-level LLM enhancement [19] injects LLM cues at multiple graph scales—yet none learns an RL filtering policy conditioned on those cues. A further line bounds how durable any text-derived cue can be: CAMERA [20] studies *semantic* camouflage, in which fraudsters deliberately mimic benign writing style. Text-based signals are the pathway that camouflage can erode most cheaply, and we return to this limitation in Section 5.

2.4 Research Gap

Each line of work addresses one mode of reasoning that camouflage-resistant fraud detection requires. GNN-based methods provide structural reasoning but lack adaptive per-node filtering and external signal. RL-guided methods add adaptive filtering but use structural-only state representations. LLM-integration work adds semantic signal but lacks a learned filtering policy. To the best of our knowledge, no prior framework unifies all three: existing methods pair at most two—GNN+RL (CARE-GNN, RioGNN, FraudGNN-RL) or GNN+LLM (FLAG, DGP). The present paper addresses this gap by routing LLM-derived signal into *both* the node features *and* a learned per-node RL filtering policy over a multi-relation backbone, trained end-to-end under a single objective.

3 The Three-Pillar Framework

3.1 Problem Setup and Notation

We model a review platform as a multi-relation graph $G = (\mathcal{V}, X, \{\mathcal{E}_r\}_{r=1}^{R})$. Each node $v \in \mathcal{V}$ is an entity—a review, in our experiments—carrying a feature vector $x_v \in \mathbb{R}^d$, and each relation r links nodes that share a particular kind of interaction; on YelpChi the three relations connect reviews by common user, by common product and rating, and by common product within a month. We write $\mathcal{N}_r(v)$ for the neighbours of v under relation r, and the task is node-level: decide whether v is fraudulent. What makes this hard is camouflage—a fraudulent node deliberately wires itself to benign neighbours, so a plain average over $\mathcal{N}_r(v)$ washes out the very signal we are trying to read.

Our framework keeps the multi-relation backbone of CARE-GNN intact and adds two things on top of it: a learned policy that decides, per node and per relation, which neighbours to trust, and a language model that supplies evidence the behavioural features never see. Figure 1 sketches how the three pillars connect; the rest of this section takes them one at a time.

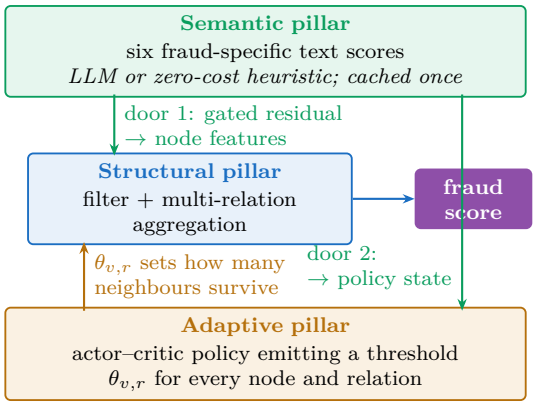


Figure 1: The three-pillar framework. The semantic pillar enters through two doors: a gated residual into the node features, and the risk scores into the policy state. Both are active in the full configuration of Section 4.2; what each contributes is measured there, and an LLM-derived initialisation of the policy, present in our original design, is removed throughout.

3.2 Multi-Relation Backbone

The backbone follows CARE-GNN [3]. A small label predictor maps each node to a class posterior $\hat{p}_v = \mathrm{softmax}(\mathrm{MLP}_\ell(x_v))$, and candidate neighbours are ranked by the label-aware distance

$$D_r(v,u) = \big|\, \hat{p}_v[1] - \hat{p}_u[1] \,\big|, \qquad u \in \mathcal{N}_r(v), \tag{1}$$

so a neighbour counts as "close" when the predictor judges it to share v's fraud likelihood rather than merely its raw features—precisely the comparison camouflage tries to defeat. Within each relation we keep the closest neighbours, aggregate their transformed features into a summary h_v^r, and then fuse the per-relation summaries into one embedding

$$h_v = \sigma\Big(W\big[\, x_v \,\|\, \textstyle\sum_{r=1}^{R} \omega_r\, h_v^r \,\big]\Big), \tag{2}$$

where ω_r weights each relation's contribution. In CARE-GNN both the number of neighbours kept and the weights ω_r are driven by a heuristic bandit; replacing that bandit is the job of the second pillar.

3.3 Camouflage-Aware Policy Network

State. Before it can decide anything, the policy needs a compact read on each node's local situation. For node v under relation r we assemble

$$s_{v,r} = \big[\, x_v \,\|\, c_v \,\|\, \deg_v \,\|\, \bar{d}_{v,r} \,\|\, j_{v,r} \,\|\, \nu_{v,r} \,\|\, z_v \,\big], \tag{3}$$

which stacks the node features; the label predictor's confidence $c_v = 1 - \mathrm{H}(\hat{p}_v)/\log 2$; four inexpensive structural statistics—normalised degree, mean label-aware distance to neighbours $\bar{d}$, Jaccard overlap j between the relation and the homogeneous neighbourhood, and neighbour feature variance ν; and, when the semantic pillar is active, the LLM risk scores z_v of Section 3.4. The features enter detached on purpose, so the policy gradient shapes the policy instead of leaking back into the embedding table.

Policy. Each relation owns a head that turns this state into the two parameters of a Beta distribution and samples a threshold from it,

$$(\alpha_{v,r}, \beta_{v,r}) = g_\psi^r(s_{v,r}), \qquad \theta_{v,r} \sim \mathrm{Beta}(\alpha_{v,r}, \beta_{v,r}), \tag{4}$$

with $\alpha, \beta = \mathrm{softplus}(\cdot) + \frac{1}{2}$ to keep them positive. At evaluation we use the mean $\theta_{v,r} = \alpha/(\alpha+\beta)$; during training we sample, which gives the policy room to explore. The scalar $\theta_{v,r}$ then does double duty. It sets how many neighbours v keeps in relation r—we retain the $\lceil \theta_{v,r}\, |\mathcal{N}_r(v)| \rceil$ closest by D_r—and its per-relation mean becomes the inter-relation weight ω_r

in Eq. (2). One learned quantity therefore controls both the filtering and the relation mixing, adapting the whole aggregation node by node rather than once per relation (Figure 2).

Learning the policy. We train θ with an actor–critic scheme. After each batch a shaped reward grades the choices the policy just made,

$$\begin{aligned} R = w_1 \,\mathrm{clip}\Big(\tfrac{\Delta\bar{d}}{|\bar{d}_{\mathrm{prev}}|+\epsilon}, -1, 1\Big) \\ + w_2\,(\mathrm{acc} - \overline{\mathrm{acc}}) \\ - w_3\,\overline{|\theta - \tfrac{1}{2}|}, \end{aligned} \tag{5}$$

rewarding it for tightening neighbourhoods ($\Delta\bar{d} > 0$) and for lifting batch accuracy above a running baseline, while gently discouraging thresholds that collapse to the uninformative 0.5. A critic $V_\eta(s_{v,r})$ estimates the expected reward; the policy is then updated by the advantage $A_{v,r} = R - V_\eta(s_{v,r})$ through the log-likelihood of the sampled threshold, and the critic is regressed onto R:

$$\begin{aligned} \mathcal{L}_{\mathrm{policy}} &= -\sum_{v,r} A_{v,r}\,\log \pi_\psi(\theta_{v,r} \mid s_{v,r}), \\ \mathcal{L}_{\mathrm{critic}} &= \big(V_\eta(s_{v,r}) - R\big)^2. \end{aligned} \tag{6}$$

Forming the advantage per (v, r) pair, rather than once per batch, is what made training stable enough to converge on YelpChi—a single batch-level REINFORCE signal was simply too noisy. We also keep the policy switched off for the first few epochs and then ramp its weight in linearly, letting the backbone settle before the policy starts rewiring its neighbourhoods.

3.4 Semantic Pillar: Language-Model Signal

The third pillar brings in evidence the behavioural features cannot express, and it enters through two separate doors. Through the first, a small, low-cost instruction-tuned language model (Claude Haiku 4.5) reads a precomputed structural profile of each node—degrees, neighbourhood overlaps, feature anomalies and similar summaries—and returns a short vector of risk scores; these are the z_v that join the policy state in Eq. (3). Through the second, where the raw review text is available, the same model scores each review on six fraud-relevant dimensions, all in $[0, 1]$:

- **generic language** — vague, templated phrasing over specifics;
- **sentiment mismatch** — review tone contradicting the star rating;
- **promotional tone** — marketing register and unbalanced superlatives;
- **copy-paste signal** — internal token repetition and low lexical variety;
- **detail authenticity** — concrete specifics (names, numbers, dates), inverted;
- **behavioural anomaly** — capitalisation, emoji and character-level repetition.

Each dimension is defined operationally enough that a deterministic scorer can compute it from surface statistics at zero cost—generic-phrase lookups, token-type ratios, positive-term density, and so on. We release both scorers; Section 4.3 measures what the language model adds over the deterministic one.

Injecting those textual scores the obvious way—concatenating them onto x_v—actually hurts, because the wider weight matrices absorb noise faster than signal. We instead fold them back through a gated residual that leaves the feature dimension untouched,

$$\tilde{x}_v = x_v + \sigma(g) \odot (W_t\, t_v), \tag{7}$$

where t_v is the textual-score vector and g is a per-dimension gate initialised at -2 so that $\sigma(g) \approx 0.12$. Training therefore starts almost exactly where CARE-GNN would, and the network opens each gate dimension only where the text earns its place—and, just as importantly, can close it again where the text is noise. Every LLM output is computed once and cached, so none of this adds anything to the per-epoch training cost. The six score definitions are deliberately interpretable, which makes the scorer itself swappable: a deterministic heuristic can compute the same six quantities at zero cost, and Section 4.3 measures exactly what the LLM adds over that baseline.

3.5 Training Objective

The three pillars share a single objective. The supervised part is the GNN classification loss together with the auxiliary label-predictor loss that also feeds the policy state; the policy and critic terms train CAPN:

$$\mathcal{L} = \mathcal{L}_{\mathrm{gnn}} + \lambda_1 \mathcal{L}_{\mathrm{label}} + \lambda_\pi \mathcal{L}_{\mathrm{policy}} + \lambda_c \mathcal{L}_{\mathrm{critic}}. \tag{8}$$

Here λ_π follows the warmup-and-ramp schedule above while the other weights are fixed. Everything is optimised end to end, so the gradients that sharpen the classifier and those that shape the filtering policy are applied together rather than in alternation.

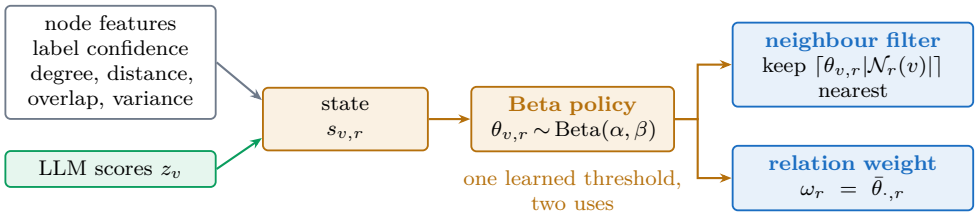


Figure 2: The CAPN mechanism. A per-node state—node features, label confidence, four structural statistics, and the LLM scores z_v—parameterises a Beta distribution whose sample $\theta_{v,r}$ does double duty: it sets how many neighbours survive the filter, and its per-relation mean becomes the relation weight ω_r of Eq. (2). The z_v channel is active in the full configuration; Section 4.2 measures what it adds.

4 Experiments

Our experiments answer two questions in order. First, the framework question: does unifying structural, adaptive, and semantic reasoning improve fraud detection over the structural base alone—and which pillar drives the gain? We answer it with a pillar-wise ablation on two variants of the CARE-GNN backbone. Second, the generality question: how far does the framework carry beyond the backbone it was built on? We answer that in two parts—by adding its semantic signal, in the simplest possible way, to eight classifiers that have nothing else to do with CAPN, and by attaching the entire framework, unchanged, to a second selection-based backbone.

4.1 Setup

We evaluate on YelpChi [21], packaged as a multi-relation graph by [3]: a review graph of 45,954 reviews (14.5% fraudulent) with 32 handcrafted behavioural features and three relations (same-user, same-product-and-rating, same-product-within-a-month). Every number in this section uses one fixed two-stage stratified split—25% train, 15% validation, 60% test. We do *not* adopt the 40%–70% training ratios of recent spectral-GNN papers—supervision ratio dominates absolute AUC on this benchmark (GAGA moves from 0.944 at 40% train to 0.70 at 1% [14])—so *every* baseline in Table 3 is re-run in-repo at this identical split rather than quoted from its source paper.

Unless noted, we report test-set AUC (and, for the framework, AP and F1-macro) at the checkpoint with the best validation AUC, averaged over five seeds (72–76) with sample standard deviation. Significance for five-seed models is a paired t-test across seeds; for the deterministic models (logistic regression, XGBoost) it is a 95% paired bootstrap confidence interval over test nodes. The language model is Claude Haiku 4.5 at temperature 0; its six textual-risk scores per review ("text6"—generic phrasing, sentiment–rating contradiction, copy-paste repetition, and similar cues) are computed once and cached, so nothing in the loop calls the external model. Our BWGNN and XGB-Graph baselines are our own implementations; we validated the BWGNN reimplementation at the original 40/20/40 protocol and obtained AUC 0.897 (Hetero) and 0.850 (Homo), matching the published 0.905/0.840 [12] closely enough to serve as a fair reference.

4.2 Main result: the unified framework

Table 1 builds the framework on the CARE-GNN backbone one pillar at a time and reports every configuration twice: on the five exploration seeds used throughout development, and—for every headline claim—on twenty *held-out* seeds never used in any tuning decision. The structural backbone alone (R1) reaches 0.7639 on the held-out set. The adaptive pillar (R2) adds +0.58 pp (positive on 20/20 seeds, $p < 10^{-4}$); the semantic pillar (R3) adds +1.97 pp (20/20, $p < 10^{-12}$); and the full framework (R4) reaches 0.7909, +2.70 pp over the backbone (20/20, $p < 10^{-16}$). Critically, the pillars *compose*: on top of the semantic signal the learned policy adds a further +0.74 pp (19/20, $p < 10^{-6}$), and on top of the policy the semantic signal adds +2.12 pp (20/20, $p < 10^{-15}$). Every pillar, and both stacking directions, is individually significant under held-out evaluation; Figure 3 shows the per-seed consistency behind those averages.

Average precision moves with AUC throughout, and more sharply—+1.90, +3.62 and +5.11 pp for R2, R3 and R4 over the backbone (all 20/20 seeds, $p < 10^{-9}$)—which is the more informative view at 14.5% positives. F_1-macro, by contrast, is flat in

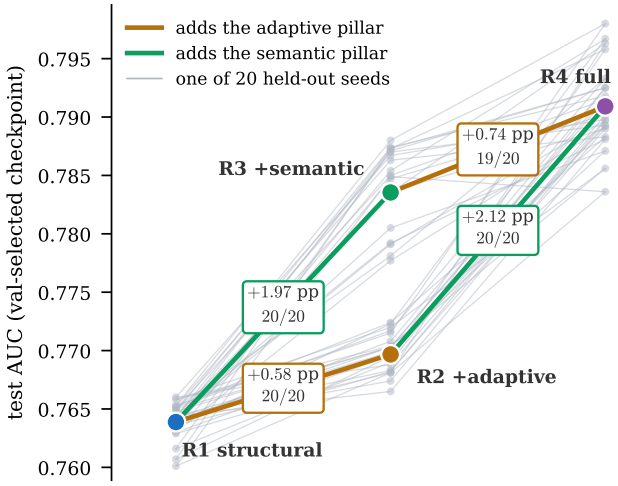


Figure 3: The held-out ablation as the 2×2 lattice it is: adaptive pillar on/off $\times$ semantic pillar on/off. Grey lines trace each of the twenty held-out seeds (100–119) along both routes from the structural base to the full framework; coloured edges are the means. Every edge isolates exactly one pillar—orange adds the adaptive pillar, green the semantic one—and carries its paired delta and the number of seeds improved. That both routes rise along their whole length is the visual form of the composition result: neither pillar's gain is absorbed by the other. Note the truncated vertical axis: the full span shown is 3.5 pp of AUC.

every comparison we ran ($|\Delta| \leq 1.3$ pp, $p > 0.3$, roughly half the seeds positive in each direction). The pillars reorder the score distribution without moving the particular operating point a fixed 0.5 cut selects; we report F_1 for completeness and rest no claim on it.

Table 1: Pillar-wise ablation on the CARE-GNN backbone, on the twenty held-out seeds (100–119) never used in any tuning decision; mean$_{\pm \text{sample std}}$. Configurations: **R1** backbone alone; **R2** adds the policy (neutral initialisation, no LLM anywhere); **R3** adds the text gate of Eq. (7); **R4** has both, and retains the LLM scores in the policy state. The LLM prior initialisation is removed throughout. On the five exploration seeds (72–76) the same ordering holds (AUC 0.7648, 0.7688, 0.7817, 0.7909). Paired deltas and p-values are given in the text.

Configuration	AUC	AP	F_1-macro
R1 CARE-GNN	$0.7639_{\pm.0016}$	$0.3721_{\pm.0056}$	$0.5968_{\pm.0244}$
R2 + Adaptive	$0.7697_{\pm.0016}$	$0.3911_{\pm.0048}$	$0.5991_{\pm.0225}$
R3 + Semantic	$0.7835_{\pm.0043}$	$0.4083_{\pm.0109}$	$0.5912_{\pm.0556}$
R4 Full	$\mathbf{0.7909}_{\pm.0038}$	$\mathbf{0.4232}_{\pm.0099}$	$\mathbf{0.6043}_{\pm.0429}$

Attributing the gain pillar by pillar requires that each configuration measure its pillar and nothing else. Our original design coupled the pillars at two points—the policy was initialised from LLM-derived priors, and LLM scores entered the policy state—and both had to be removed before the adaptive pillar could be isolated. Doing so reverses the conclusion: prior-initialised, the adaptive contribution reads as a null (+0.19 pp, $p = 0.10$); neutrally initialised, it is +0.58 pp on 20/20 held-out seeds. The priors *suppress* the policy they were meant to assist, and the scores-in-state coupling contributes nothing measurable (Table 2). These are the clean configurations Table 1 reports. Beyond attribution hygiene this is a design finding in its own right: hand-designed couplings between components can suppress the very component they are meant to help, and only a deconfounded ablation exposes it.

Does this survive a different split? Twenty seeds on one partition measure reproducibility under re-initialisation, not generalisation: every configuration above shares the same train/validation/test division, so the p-values quantify the wrong variance component for the claim we want to make. We therefore re-ran R1 and R4 on ten independently drawn stratified splits. The framework improves the backbone on 10/10 of them, by +1.80 pp AUC (95% CI $[+1.40, +2.20]$, $p = 3 \times 10^{-6}$) and +2.24 pp AP. The effect is real but smaller than the frozen split suggests, and the two variance components differ by more than a factor of two: the backbone's AUC varies by ± 0.0016 across seeds and ± 0.0038 across splits.

The same pattern holds on Amazon, where the adaptive pillar gives +0.61 pp across ten splits (10/10, $p = 1.5 \times 10^{-5}$) against ±0.0027 seed and ±0.0075 split variation. We report the split-level figure as the headline because it is the one the generalisation claim rests on—and because of what the individual splits show. On YelpChi the frozen split turns out to be the most favourable of the ten (+2.46 pp against a +1.80 pp mean); on Amazon it is below average (eighth of ten). The split was fixed before any of these runs and not chosen on that basis, but a single split cannot be known to be representative in advance, and here one of ours was not. That is precisely the argument for reporting the split-level estimate rather than the frozen-split one.

Table 2: What the two cross-pillar couplings cost. Both were part of the original design; both are removed in the configurations of Table 1.

Coupling	Effect	p
LLM priors → policy init, alone	−0.21	—
LLM priors → policy init, in R4	−0.71	—
LLM scores → policy state (on removal)	+0.33	0.44

4.3 Generality I: the semantic pillar beyond the framework

If the semantic pillar is the framework's strongest pathway, a natural question is whether that strength is bound to our architecture or is a property of the signal itself. Table 3 answers it by adding text6 to eight classifiers of increasing strength, from a linear model to XGB-Graph—the tree-plus-neighbour-aggregation method that tops YelpChi on the GADBench benchmark [15]. The injection is plain concatenation (only CARE-GNN uses the gate of Eq. (7)), and the table isolates the *signal*: the adaptive pillar is not applied here.

The signal transfers across every model family, including the strongest backbone we have: XGB-Graph rises 0.9106 → 0.9177 AUC (AP 0.720 → 0.727). The gain is larger on weaker backbones (Figure 4) but remains positive—and usually significant—everywhere, including at the top of the range. We deliberately do not quote a slope for that relationship. Two things make a fitted rate untrustworthy here: the gain is a difference that contains the baseline on which it is regressed, which biases any such slope negative even under the null; and the fit is dominated by one point, GraphConsis, whose removal moves the least-squares slope from −2.2 to −0.8 pp per +0.1 AUC. The qualitative statement is robust to both objections; a quantified rate is not. The semantic pillar is therefore not an architecture-bound component but a portable source of fraud signal the behavioural features do not already contain.

Table 3: The framework's LLM textual-risk signal (text6) added to eight backbones at the identical YelpChi 25/15/60 split. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ (paired t-test over seeds, or 95% bootstrap CI excluding zero for the deterministic models); † not significant.

Backbone	AUC	+ text6	Δ
Feature-only classifiers			
Logistic regression	0.7694	0.7923	+2.29*
MLP (64,64)	0.8205	0.8283	+0.78†
HistGBM	0.8620	0.8738	+1.18***
XGBoost	0.8541	0.8783	+2.42*
Graph-aware models			
CARE-GNN (gated)	0.7648	0.7817	+1.69***
BWGNN-Homo	0.8252	0.8417	+1.66**
BWGNN-Hetero	0.8879	0.8941	+0.62***
XGB-Graph (2-hop)	0.9106	0.9177	+0.70*

Does the gain belong to the gated injection or to the signal? On BWGNN-Hetero, concatenation and the gate perform identically (+0.65 vs. +0.59 pp, difference $p = 0.62$): the gate is a *safe* injection—never worse than concatenation, and materially better only on fragile backbones like CARE-GNN, where concatenation dilutes the signal and the gate's near-closed initialisation protects early training.

A sharper question: does the signal require a language model at all? The six score definitions are interpretable enough to admit a deterministic, zero-cost heuristic scorer, and the head-to-head (Table 4) shows that although the two scorers produce genuinely different vectors (mean $|\Delta| = 0.20$; per-column correlation −0.16 to 0.56), their downstream AUC is statistically indistinguishable on *every graph backbone*. The LLM's small edge survives only on feature-only classifiers, where no aggregation smooths it away. The pillar's value lies in *which six quantities are measured*, not in who computes them.

The pillar's structural boundary is Amazon: that benchmark ships no review text, so the textual pathway cannot exist, and the only available graph-derived LLM scores add essentially nothing to its al-

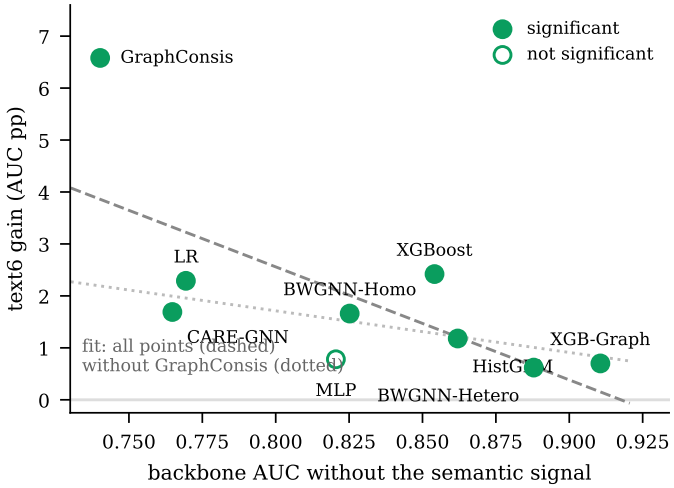


Figure 4: Dose–response of the semantic signal: AUC gain from text6 vs. the backbone AUC without it (filled = significant). Two fits are shown: over all nine points (dashed) and excluding GraphConsis (dotted), whose leverage moves the slope by a factor of nearly three.

ready strong features (+0.05 pp in the policy state; −0.59 pp as node features). This is the dose–response taken to its limit—the semantic pillar helps where text exists and features leave headroom—so we evaluate the textual pathway on YelpChi and state that scope explicitly.

4.4 Generality II: the framework as a backbone-agnostic wrapper

Does richer neighbour selection give the policy more to contribute? We test this by attaching the identical CAPN machinery—policy, critic, shaped reward, and gate, unchanged—to GraphConsis [9], whose native selection filters neighbours by feature consistency against a global threshold. Only two elements are backbone-specific: the policy state and the actuation point (a per-node consistency threshold replacing the global one). Our reimplementation reaches 0.835 ± 0.023 at the original 40/20/40 protocol—well above published reproductions (0.62–0.70)—so the wrapper is tested on a strong host, not a strawman.

Table 5 gives the result: every pathway transfers, and the complete framework improves the host by +8.61 pp ($p < 10^{-4}$). The semantic pillar carries the larger share (+6.58 pp, 18/20 seeds, 95% CI $[+3.07, +10.08]$) and the adaptive pillar is significant here too (+2.68 pp, $p = 0.015$)—a larger effect than on CARE-GNN, where it is +0.58 pp. Stacked on the semantic signal the policy adds a further +2.03 pp, positive on 13/20 seeds but not significant ($p = 0.15$).

Two things deserve comment. First, the mechanism is visible per seed. GraphConsis is volatile—its standalone AUC ranges from 0.600 to 0.800 across the twenty seeds—and the semantic signal does most of its work on the bad draws, lifting seed 106 from 0.600 to 0.838 and seed 104 from 0.669 to 0.833. That is the dose–response of Section 4.3 taken to its limit: this is the weakest host we tested and it shows the largest gain, so we read +8.61 pp as evidence that the framework *generalises* rather than as a headline accuracy result. The configuration still reaches only 0.826.

Second, an earlier five-seed version of this experiment reported all three comparisons as non-significant (+1.20 pp at $p = 0.63$ for the policy, +2.71 pp at $p = 0.38$ for the framework), and we drew the wrong conclusion from it: that the adaptive effect was real but below this host's detection floor, requiring $n \approx 340$ seeds to resolve. Twenty seeds resolved it.

Table 4: Does the semantic signal need the LLM? Haiku-scored minus heuristic-scored text6, identical split and protocol. Significance is a paired t-test over seeds, or a 95% paired bootstrap CI over test nodes for the deterministic models (logistic regression, XGBoost), which are bit-identical across seeds. $^{*}p < 0.05$, $^{**}p < 0.01$; † not significant.

Feature-only	Δ (pp)	Graph-based	Δ (pp)
Logistic reg.	$+0.42^{*}$	CARE-GNN	$+0.15^{\dagger}$
MLP (64,64)	$+0.47^{\dagger}$	BWGNN-Hetero	$+0.14^{\dagger}$
HistGBM	$+0.72^{**}$	BWGNN-Homo	$+0.70^{\dagger}$
XGBoost	$+0.55^{*}$	GraphConsis	$+1.22^{\dagger}$

Table 5: Standalone vs. framework-integrated GraphConsis, twenty seeds (100–119) at the frozen 25/15/60 split, deltas paired seed for seed. An earlier five-seed version of this table reported every comparison as non-significant; the discrepancy is discussed in the text.

Configuration	AUC	Δ vs. standalone	p
GraphConsis	0.7402 ± 0.0444	—	—
+ adaptive	0.7670 ± 0.0337	+2.68	0.015
+ semantic	0.8060 ± 0.0519	+6.58	0.0009
+ full framework	$\mathbf{0.8263 \pm 0.0337}$	**+8.61**	$< \mathbf{10^{-4}}$

The five-seed standalone baseline was also optimistic by 4.4 pp (0.7840 against 0.7402). We note this because the same five seeds supported a further claim we now withdraw: that the semantic signal reduces this host's seed variance by 3.5× (0.0357 → 0.0100). At twenty seeds it does not (0.0444 → 0.0519; variance ratio $F = 0.73$, $p = 0.50$). A variance ratio estimated at four degrees of freedom carries a confidence interval spanning an order of magnitude. On a backbone this volatile, five seeds was simply too few for any of these quantities, in either direction.

The wrapper experiment thus establishes portability while completing the attribution. The framework transfers with two localised changes, and both pathways survive the move: the semantic one at +1.97 pp on CARE-GNN and +6.58 pp here, the adaptive one at +0.58 pp, +0.51 pp and +2.68 pp across CARE-GNN on YelpChi, CARE-GNN on Amazon, and GraphConsis respectively. The adaptive pillar is therefore significant on three hosts rather than confined to the one it was developed on.

5 Conclusion

We introduced a framework that unifies structural, adaptive, and semantic reasoning—GNN, RL, and LLM—as three pillars of camouflage-resistant fraud detection, trained under a single objective, and we measured what each pillar contributes rather than assuming it. The framework delivers: it outperforms its structural base by +1.80 pp across ten independent data splits (10/10 positive, $p = 3 \times 10^{-6}$; +2.70 pp on twenty seeds at the frozen split), transfers to a second selection-based backbone with only its state features and actuation point rewritten, and its semantic pillar proves portable well beyond the framework itself—significantly improving seven of the eight backbones we tested, including the benchmark-leading tree-aggregation method. Attached whole to a second selection-based backbone it adds +8.61 pp ($p < 10^{-4}$). The pillar-wise attribution is equally clear and we state it plainly: the semantic pillar contributes the largest gain; the adaptive pillar contributes a smaller one whose clean measurement required de-confounding—the LLM-derived prior initialisation suppresses the policy, and removing it, followed by twenty-seed held-out confirmations, establishes the pillar's true effects: +0.58 pp alone and +0.74 pp composed on top of the semantic pillar (19/20 seeds, $p < 10^{-6}$), replicating at +0.51 pp on Amazon and +2.68 pp on GraphConsis. Every pillar thus carries a rigorously established, held-out-confirmed contribution on three hosts, and the two learned pillars compose rather than substitute. A unified framework whose components can be measured, detached, and re-hosted is precisely what makes that accounting possible—including the detection of a suppressive coupling in our own design, and the correction of three numbers that five seeds had estimated wrongly. Demonstrating a gain at a deployable operating point, rather than in ranking quality alone, is the framework's next step.

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